



Ou Ocean Kun Hei

HKUST(GZ)

BSc in Data Science and Big Data Technology
(Expected May 2027)

+86 15112134690

✉ okhou533@connect.hkust-gz.edu.cn

🌐 <https://github.com/oukunhei>

Education

Hong Kong University of Science and Technology (Guangzhou)	2023 – Present
Relevant Coursework: Cloud Computing and Big Data Systems, Deep Learning, Mathematics for Data Science, Introduction to Natural Language Processing, Database Management Systems, Machine Learning, Discrete Mathematics, Applied Statistics	CGPA: 3.340/4.3

Research Experience

Incremental Semantic Graph Construction for Data Agents in Heterogeneous Data Environments (Final Year Project)	Aug 2026 – Present
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- **Overview:** Investigating how data agents can extract, validate, and accumulate database semantics from execution traces as persistent graph-structured memory. Designing an incremental semantic graph framework and evaluating it against trajectory, text, vector, and static-schema memory in terms of task accuracy, knowledge quality, and execution efficiency.

DPC: Training-Free Text-to-SQL Candidate Selection via Dual-Paradigm Consistency (ACL 2026) 🔗	Sep 2025 – Jan 2026
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- Developed **DPC (Dual-Paradigm Consistency)**, a training-free SQL candidate selection framework that constructs **Minimal Distinguishing Databases (MDDs)** and cross-validates candidate behavior through complementary SQL and Python execution, improving selection accuracy by up to **2.2%** over Self-Consistency baselines.
- As the second author, I was primarily responsible for experiments across multiple models and systems, case analysis, and paper revision and writing. The work has been accepted by ACL 2026.

SQL Candidate Selection via SFT and Preference Optimization 🔗	Dec 2025
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- Developed a learned SQL candidate selector for Text-to-SQL on the BIRD benchmark, using database schemas, natural-language questions, and candidate SQL queries generated by multiple Text-to-SQL systems as model inputs.
- Built an end-to-end training and evaluation pipeline based on Qwen2.5-Coder, including LoRA-based supervised fine-tuning (SFT), Direct Preference Optimization (DPO) and execution-based validation against database results.

Internship

Lumotech, Development Engineer	May 2026 - Jun 2026
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- Developed the LLM agent module for an AI-powered interior design platform, designing reusable agent skills and tool-calling workflows for dialogue-driven task execution.

Skills and Research Interests

Programming: Python, SQL; working knowledge of C++

Data & Database: Relational databases, data analysis with Python/Pandas

LLM & Machine Learning: PyTorch, Transformer architectures, RAG, LoRA/SFT/DPO, tool-using LLM agents

Systems: Linux development, cloud computing, reproducible experiment pipelines

Research Interests: Data-centric AI, LLM-based Data Agents, NL2SQL

Honors & Awards

Second-Class Scholarship for Hong Kong, Macao, and Overseas Chinese Students	2025
Merit Research Award of Undergraduate Research Program	2024
HKUST(GZ) First Class Scholarship	2023